



Editorial

The prosody–syntax interface: Focus, phrasing, language evolution

1. Introduction

This issue presents collected papers from the second prosody–syntax interface workshop (PSI 2), held in Berlin in June 2008. The workshop was presented by the Center for General Linguistics (ZAS, Berlin) and the Collaborative Research Center 632 “Information Structure” (SFB 632, Berlin/Potsdam), and organized by (in alphabetical order) Nicole Dehé, Laura J. Downing, Ingo Feldhausen, Shinichiro Ishihara, and Franziska Scholz.¹ The goal of the workshop was to bring together cutting-edge research by international scholars investigating the interface between prosody and syntax from a variety of theoretical perspectives, looking at a diverse range of languages and language families. The present special issue has the same goal and comprises 8 out of the 12 papers presented at PSI 2. Some of these articles provide new insights into topics which have already been extensively discussed in the literature; others introduce new perspectives. We hope that this special issue will raise further interest in this fascinating field of research and will stimulate further discussion.

In research at the prosody–syntax interface, two areas stand out as arguably the most essential: *focus* and *prosodic phrasing*. They also play a central role in the present issue. The first two papers (Ishihara; Kabagema-Bilan et al.) discuss issues related to focus and its prosodic realization. The next five papers address issues related to prosodic phrasing; the languages discussed include German (Féry), Catalan (Prieto; Feldhausen), and the Bantu languages Chichewa (Downing and Mtenje) and Chimwiini (Kisseberth and Abasheikh). The eighth and final paper in this special issue, by Sandler et al., introduces a third topic: language evolution. It is concerned with a rather unique but important subject, namely the development of the complexity of the prosody–syntax interface in a newly born language: the sign language Al-Sayyid Bedouin emerging in Israel.

In Section 2 of this editorial, fundamentals of the prosody–syntax interface will be introduced, including the notions of focus and prosodic phrasing. Section 3 summarizes the eight papers that comprise this issue and highlights their particular contributions to the field. Section 4 concludes with some general remarks.

2. The prosody–syntax interface and the notions of focus and prosodic phrasing

The results of previous research strongly support the assumption of a close relation between prosodic and syntactic structure. In the early generative framework, phonological structures were studied in terms of a linearly organized sequence of segments, or bundles of distinctive features (e.g., Chomsky and Halle, 1968). However, since the birth of the autosegmental model (generally ascribed to Pierrehumbert, 1980), along with important publications on prosodic constituency (e.g., Nespor and Vogel, 1986, 2007; Selkirk, 1984, 1986), and with the development of metrical theory (Hayes, 1995), prosody has been considered to have its own hierarchical organization (*prosodic structure*) independent of syntactic structure. The prosodic structure of an utterance is the way it is organized into prosodic constituents/categories. Above the level of the word, the following phrasal levels have generally been assumed: the phonological and/or intermediate phrase, the intonational phrase, and the utterance. Other categories (or names for the same categories) also found in the literature include the clitic group, the accentual phrase, and the minor and major phrase (see, e.g., Pierrehumbert and Beckman, 1988; Selkirk and Tateishi, 1988). Here we will use *prosodic phrases* as a cover term for all these constituents, following Ladd (2008:288). One major difference between prosodic and syntactic structure has been spelled out in terms of the *Strict Layer Hypothesis* (SLH; Selkirk, 1984), which comprises a set of structural limitations on the prosodic hierarchy. The SLH holds that prosodic categories are hierarchically ordered but non-recursive. Consequently, a prosodic constituent cannot dominate another prosodic constituent of the same type. More recent research, however, argues that prosodic structure can indeed be recursive (Féry,

¹ For further details on PSI 2, see <http://linguistlist.org/issues/19/19-1604.html>. The first PSI workshop took place in London in October 2006, run by the Centre for Human Communication, University College London; cf. <http://linguistlist.org/issues/17/17-2743.html>.

2010a; Itô and Mester, 2009, forthcoming; Kisseberth, 2010; Ladd, 1986, 1988, 2008:#8.2.4; Wagner, 2005, among others), an idea that is also discussed in the present special issue (Féry; Ishihara; Kisseberth and Abasheikh).

Since the beginning of research on prosodic structure in the present framework in the 1980s and the early 1990s, many studies have been carried out, and various proposals have been made. Two lines of research, or broad research questions, are particularly relevant in the present context: (1) Comparative prosody/prosodic typology: How universal, or how cross-linguistically variable, are the SLH and other theoretical assumptions in prosodic phonology? (2) Syntax–prosody mapping: How are prosodic and syntactic structures related to each other, and how can mismatches between them be explained?

Cross-linguistic comparisons among languages and language families have recently been addressed more systematically, resulting in comparative work in prosodic phonology (e.g., Jun, 2005, forthcoming). The present issue makes a contribution to this cross-linguistic goal, as it includes studies on a wide range of languages and language families (Germanic, Romance, Bantu, Mandarin Chinese, Japanese, and a newly emerging sign language).

The current special issue also contributes to an understanding of the syntax–prosody mapping, since it deals in great detail with prosodic phrasing, an area known for its important role. Boundaries between prosodic phrases may be identified on the basis of phonetic correlates (e.g., F0 cues, pause duration, preboundary lengthening). Various factors have been shown to influence the placement of prosodic phrase boundaries. Syntactic constituency is generally taken as a fundamental factor, an emphasis that has led to prominent approaches to syntax–prosody mapping, such as Selkirk's (1984, 1986, 1995) edge-based account, Selkirk's (2009) match theory, Nespor and Vogel's (1986, 2007) relation-based account, and Truckenbrodt's (1999) wrapping theory; see Shattuck-Hufnagel and Turk (1996) for an overview. However, it has also been shown that non-syntactic factors such as speech rate, constituent weight and length, style of speech, and semantic considerations also influence the size and shape of prosodic constituents (see Nespor and Vogel, 1986, 2007; Zec and Inkelas, 1990; Ghini, 1993; Selkirk, 2000; Sandalo and Truckenbrodt, 2002; D'Imperio et al., 2005; Dehé, 2009, among others). While earlier scholars assume that juncture constitutes an independent phenomenon of intonation (cf. Trager and Smith, 1951; Halliday, 1967), Ladd (2008) points out that this phenomenon should be considered a consequence of the prosodic structure, while pitch and relative prominence are the two phenomena that are the central aspects of intonation: “intonational features of pitch and relative prominence are distributed in utterances in ways *allowed by the prosodic structure*” (Ladd, 2008:9; emphasis in original).

The term (*relative*) *prominence* has often been linked with the concept of focus. The term *focus* originated around 1970 (cf. Halliday, 1967, or, within the generative tradition, Chomsky, 1971, and Jackendoff, 1972) and roughly refers to “the information in the sentence that is assumed by the speaker not to be shared by him and the hearer” (Jackendoff, 1972:230) or, as Halliday (1967:202) puts it, focus “indicate[s] what new information is being contributed”. Since then, research on focus – its meaning as well as its manifestation in the prosodic system (in terms of, e.g., prominence) – has received a lot of attention (cf. Krifka, 2008; Ladd, 2008:ch. 6; and Féry, 2011, for useful overviews).

Despite the fact that focus has been studied for several decades and by many researchers, even providing an adequate definition is not an easy task. Some researchers treat focus as a single, universal grammatical notion (e.g., Rooth, 1992), while others distinguish two or more different types of focus (e.g., É. Kiss, 1998; Selkirk, 2002, 2008). Another difficulty is that focus may be realized in many different ways within and across languages. It may be marked morphologically or in the syntax (i.e., word order), or it may be marked prosodically. The study of how focus is manifested in different languages or within languages in certain discourse situations is one of the central issues in research on the prosody–syntax interface.

In recent years, the interface between syntax and prosody has been approached from different theoretical perspectives. For example, in Optimality Theory (OT: Prince and Smolensky, 1993/2004), the SLH has been translated into a group of violable constraints, and typological differences have been explained in terms of differences in constraint ranking (Selkirk, 1995; Truckenbrodt, 1995, 1999; Samek-Lodovici, 2005, among others). Here as elsewhere, OT has been very successful for two reasons. First, the framework easily expresses cross-linguistic variability: reranking of the same set of constraints derives different types of languages (see, e.g., Truckenbrodt, 1999). Second, in OT, constraints from various components of the grammar (such as syntax, prosody, information structure) may be part of the same evaluation (see, e.g., Samek-Lodovici, 2005). Most of the papers in this issue present a detailed Optimality-Theoretic analysis. In addition, some of the papers extend the standard use of the OT framework, by integrating syntactic notions of the Minimalist framework (specifically the *phase*, from Chomsky, 2000, and subsequent work) or by using a version of OT which is able to cover frequency-dependent variation in the data (stochastic OT: Boersma and Hayes, 2001).

In the remainder of this editorial the eight papers compiled in this volume will be summarized and their contributions to the field will be highlighted.

3. Summary of the papers and their contribution to the field

3.1. Focus

Two papers in this volume are concerned with focus. In “Japanese focus prosody revisited: Freeing focus from prosodic phrasing”, Shinichiro Ishihara studies the prosodic realization of focus in Japanese, which has often been analyzed in terms of prosodic phrasing (Pierrehumbert and Beckman, 1988; Nagahara, 1994; Truckenbrodt, 1995; Sugahara, 2003, among many others). Based on empirical evidence from the literature as well as from his own experimental study, he first shows that focus does not affect phonological phrasing in Japanese. Then he proposes a new analysis in which focus instead

manipulates metrical prominence in the sentence. He introduces the idea that metrical culminativity, that is, the requirement that each prosodic domain contains one and only one metrical prominence on its head, is a violable constraint rather than a language-universal axiom. In Japanese, in which he claims the CULMINATIVITY constraint is ranked relatively low, a multi-headed prosodic phrase (a prosodic phrase containing more than one metrical prominence) and a headless prosodic phrase may both occur whenever focus requires an additional prominence on the focused element and prohibits prominence within the post-focal domain. Another higher-ranked constraint, namely the syntax–prosody mapping constraint, keeps the prosodic phrasing unchanged.

Elena Kabagema-Bilan, Beatriz López-Jiménez, and Hubert Truckenbrodt investigate the phonetic realization of “Multiple focus in Mandarin Chinese”. Based on Xu’s (1999) experimental study on single focus – which reports that a single (non-sentence-final) focus in Mandarin Chinese has F0 raising in the focus and F0 lowering and compression in the post-focal area – they examine F0 and duration of each syllable in test sentences containing multiple foci, pronounced in five different discourse contexts. The results show that the first focused element of a multiple-focus construction (specifically a two-focus construction) neither shows significant F0 raising nor triggers post-focal F0 lowering/compression, while the second focus does show both expected F0 effects (raising on the focus, lowering in the post-focal area).

New insights: Ishihara’s contribution is a new insight into the phonological interpretation of focus. In the literature, the effect of focus on the relation between syntactic and prosodic structure is a hotly debated issue. One major question concerns the phonological substance affected by focus. While many researchers adopt the idea that in some languages focus may influence prosodic phrasing (i.e., insertion and deletion of phrase boundaries, or enhancing and softening thereof),² it has also been argued, for example, that focus directly affects pitch register while keeping the phrasing intact (Féry, 2010b; Féry and Ishihara, 2010).³ Ishihara strictly separates the phonological effect of syntax–prosody mapping, which affects prosodic phrasing, and the effect of focus, which affects metrical prominence. This separation of prosodic phrasing and metrical prominence is made possible by allowing culminativity to be violated under certain circumstances.

Kabagema-Bilan et al.’s finding is an important contribution to the theory of focus and its realization. In particular, the data indicate that the requirement that the focus bears the strongest prominence only needs to be met by the second/last focused element. It has been observed for many different languages that prosodic prominence indicates focus (Jackendoff, 1972, and much subsequent work), and this relation has been given different names, among them *Focus* (Truckenbrodt, 1995), *Stress–Focus Correspondence* (Reinhart, 2006) and *Focus Prominence Rule* (Zubizarreta, 1998). However, while the relation between focus and prominence has been well established for single foci, multiple foci have rarely been addressed. The paper by Kabagema-Bilan et al. seeks to close this gap and establishes that in multiple-focus constructions consisting of two foci, only the second shows the relevant phonetic effects.

3.2. Prosodic phrasing

This section is devoted to the five contributions on prosodic phrasing. In “German sentence accents and embedded prosodic phrases,” **Caroline Féry** investigates sentence-accent assignment in German, a topic that has already been discussed by many researchers (e.g., Höhle, 1982; Krifka, 1984; Gussenhoven, 1984, 1992; Kratzer and Selkirk, 2007). She introduces recursive p-phrasing, which reflects syntactic phrasing more faithfully than non-recursive structures following the SLH, which have often been assumed in the past. Féry first explains the interaction of syntactic and prosodic factors that influence the pattern of accent assignment between a predicate and its complement – an interaction that constitutes the so-called *integration* phenomenon – using a set of OT constraints that regulate the syntax–prosody mapping. She then extends the analysis of integration to other, more puzzling cases, such as sentences with intransitive verbs, resultative predicates, or more than two lexical arguments. It is shown that while syntactic factors determine prosodic phrasing, certain prosodic factors, such as postnuclear deaccenting, crucially play a role in accent assignment. With this approach, Féry tries to integrate the discussion of certain phenomena which have been discussed separately in the past into one unified account.

In “Prosodic effects on phrasing: Clash avoidance in Catalan”, **Pilar Prieto** argues that the construction of phonological phrases in Catalan does not exclusively rely on syntactic information but also depends on aspects of prosodic well-formedness. In Catalan, strings of adjacent stressed syllables are typically avoided, a tendency which is often assumed to be universal. Stress-clash avoidance has a clear influence on the placement of prosodic boundaries. Prieto conducts a production experiment with different types of syntactic structures (SVO, NP, and VP structures) consisting of two phonological phrases the first of which ends with a target word and the second of which begins with a target word. In condition 1, the metrically strong syllables of the two target words are adjacent (stress clash), while in condition 2, the two words do not have adjacent strong syllables (no clash). Prieto shows that in the stress-clash condition the first accent involved in the clash becomes destressed/deaccented and a restructuring of the phrasing pattern takes place such that the two target words end up in one p-phrase. The results offer empirical evidence that phrasing decisions in Catalan also depend on the metrical structure of the sentence.

Ingo Feldhausen examines “The prosodic phrasing of sentential objects” in Catalan and argues that syntactic information is indispensable for the construction of higher-level prosodic categories in that language—although prosodic/eurhythmic

² Previous research has shown that focus may either soften the strength of a boundary (e.g., Cho, 1990, for Korean; Dehé, 2008, for Icelandic) or enhance it (e.g., Hayes and Lahiri, 1991, for Bengali; Kanerva, 1990, for Chichewa; Selkirk and Shen, 1990, for Shanghai Chinese; Selkirk, 2000).

³ Frota (2002) presents an analysis of European Portuguese, in which focus does not affect prosodic phrasing.

constraints play an important role, too. He shows that the phrasing of SVO structures with a sentential object allows for frequency-dependent variation, with (SV)(O) being the most common pattern, which contrasts with the default SVO phrasing pattern (S)(VO). His main research question is whether the prosodic separation of the object clause from the matrix clause is triggered by the syntactic status of the object as a full clause (i.e., CP) or by the prosodic weight of the object. The possible relevance of prosodic weight stems from previous research on Catalan SVO structures with nominal objects: (SV)(O) only occurs if the DP object is prosodically heavy, that is, if it consists of two or more prosodic words (ω). Feldhausen reports on two production experiments that respectively test prosodically heavy CP objects and prosodically light CP objects. Even though variation was attested in both experiments, the most common outcome was for the matrix S and V to phrase together and to be prosodically separated from the clausal object. Feldhausen thus concludes that the syntactic projection CP strongly influences prosodic structure, irrespective of its prosodic weight. He provides an analysis in the framework of stochastic OT (Boersma and Hayes, 2001) to account for the frequency distribution of the various phrasing patterns. Moreover, he introduces a new constraint to the ALIGNMENT family: ALIGN-CP,L. This constraint aligns the left edge of a CP with the left edge of a prosodic phrase and thus accounts for the boundary preceding the object clause.

Laura Downing and Al Mtenje's study "Un-Wrap-ing prosodic phrasing in Chichewa" argues that Truckenbrodt's (1995) WRAP constraint formalizes the wrong generalization about prosodic phrasing in Chichewa, a Bantu language which provides an important testing ground for the edge-based approach to prosodic phrasing (developed in Selkirk, 1986, and subsequent work). While the WRAP constraint demands that an XP be parsed into a single phonological phrase – which was sufficient to account for Chichewa's simpler structures – the authors show that prosodic breaks do show up in more complex XPs. For their study, they consider new, original data from Chichewa featuring nominal and verbal modifiers, which render the XPs examined larger and internally more complex. Nominal modifiers (i.e., relative clauses) are obligatorily followed by a break and verbal modifiers (i.e., verbal adjuncts) are often preceded by a break. As a consequence, WRAP is violated. Downing and Mtenje account for the phrasing pattern by proposing (a) that phonological phrases align with edges of syntactic phases (ν P and CP) rather than with edges of lexical XPs and (b) that nominal modifiers are inherently focused.

In "Chimwiini phonological phrasing revisited" **Charles Kisseberth and Mohammad Imam Abasheikh** present data that enables them to re-examine four important aspects of phrasing in Chimwiini. They conclude that the language provides unrivaled support for the basic tenets of the edge-based approach to prosodic phrasing, including its OT extension (Selkirk, 1986, 1995, 2000). First, the authors show that the accentual system is an even more robust source of evidence for phrasing in Chimwiini than penultimate vowel lengthening, which was previously argued to be the main indicator of phonological phrasing (Selkirk, 1986). Each and every phrase has just one accent, which is located on the last word in the phrase and is always phonetically observable. Second, conjoined structures, relative clauses, and sentential complements provide strong supporting evidence for the critical claim of Selkirk's indirect-reference model, namely that a phonological phrase is not identical to a syntactic phrase. Third, the authors show that alignment of the right edges of phonological phrases and XPs is a generalization that is correct and fundamental to an analysis of Chimwiini phrasing, but is insufficient at the same time, since phonological phrase breaks occur in other locations as well. By considering data from different grammatical categories and concepts (such as morphological negative verb forms, the distinction between definiteness and indefiniteness, and emphasis/focus) the authors argue that further, more specific alignment constraints are necessary to account for the phrasing patterns. Fourth, Kisseberth and Abasheikh give clear evidence that Chimwiini has recursive phrasal structure and that both WRAP-XP and ALIGN-XP,R can and have to be satisfied at the same time, since complex facts in the accentual system of Chimwiini are best analyzed in this way.

New insights: The papers by Pilar Prieto and Ingo Feldhausen concentrate on Catalan, a language which has received a lot of attention in recent years, including the recent development of the Cat_ToBI labeling system (Aguilar et al., 2009; Prieto, forthcoming). Studies such as D'Imperio et al. (2005) and Feldhausen (2010) support the assumption that in Catalan at least, syntactic and prosodic factors interact in shaping prosodic structure. The two papers in this volume continue this line of research. Prieto adds the impact of metrical structure to the factors affecting prosodic structure in Catalan, while Feldhausen's data provide evidence for the relevance of syntactic categories independent of prosodic weight.

German phrasing has also been well studied in previous work. To this, Caroline Féry adds the insight that the notion of integration – which in previous accounts refers to the integration of predicates into the prosodic phrase of their argument – extends to a larger set of cases than previously assumed. In her account, this follows from recursion of prosodic structures, or, as she puts it, "embeddedness of prosodic domains into larger ones that contain them". From this it also follows that two phrases may share the same accent. This paper thus also adds to the growing literature assuming recursion in prosodic structure.

The contributions by Laura Downing and Al Mtenje and by Charles Kisseberth and Mohammad Imam Abasheikh concentrate on the Bantu languages Chichewa and Chimwiini, respectively. Both languages have been subject to prosodic research since the 1970s and 1980s and important theoretical claims have been made based on these languages (cf. work by Kanerva, Selkirk, and Truckenbrodt, among others). The contribution of the two papers at hand is twofold: they offer new empirical data and they use this data to modulate prosodic theory. Making reference to syntactic phases (Chomsky, 2000, and subsequent work), Downing and Mtenje add a new violable constraint to the ALIGNMENT family to account for the syntax–prosody mapping in Chichewa. The syntactic projections ν P and CP are generally considered phases. At Spell-Out, ν P and CP contain the relevant information which is sent to the PF interface (cf. Hornstein et al., 2005). Following the introduction of OT into prosodic phonology in the 1990s (e.g., Selkirk, 1995), syntactic notions deriving from the Minimalist framework have in turn been introduced to prosodic phonology, with the phase and multiple Spell-Out being identified as concepts relevant to

syntax–prosody mapping (Legate, 2003; Kahnemuyipour, 2004; Adger, 2007; Ishihara, 2003, 2007; Kratzer and Selkirk, 2007, among others). Downing and Mtenje's contribution adds to this line of research.

While most of the articles in the current issue are couched in the standard OT framework, Feldhausen additionally exploits stochastic OT (Boersma and Hayes, 2001) to explain the variation in phrasing patterns, and the difference in the frequency of these patterns, found in one language. Since prosody is known (even notorious) for its variability, which makes it difficult to establish a solid theoretical model that is powerful enough but not too powerful to make the correct predictions, inter- and intra-speaker variation in phrasing constitutes one of the areas that require further research in this field.

Despite the new results and perspectives in research in prosodic phonology, the current issue also shows that previous findings are still valid and of great relevance. Kisseberth and Abasheikh present a new set of data supporting previous claims. Their data supports, for example, the assumption of indirect reference between syntactic and prosodic structure, a claim made in Selkirk (1986). They furthermore support the theory of alignment and add the need for more specific alignment constraints. Although not formulated explicitly, this desideratum is also apparent in the introduction of the specific alignment constraints ALIGNR-PHASE and ALIGNR-PHP in Downing and Mtenje's contribution and ALIGN-CP,L in Feldhausen's paper.

3.3. Language evolution

Although focus and prosodic phrasing account for the majority of studies in research on the prosody–syntax interface, there are other, relatively less studied issues as well. One such area, and a topic that receives attention in this special issue, is the development of prosodic structure and syntactic structure and of their interaction in a relatively newly emerging language. Wendy Sandler, Irit Meir, Svetlana Dachkovsky, Carol Padden, and Mark Aronoff address “The emergence of complexity in prosody and syntax” in Al-Sayyid Bedouin Sign Language (ABSL), a sign language in Israel that is still at an early stage of its development. They examine various prosodic cues for constituency and intonation (e.g., manual timing, facial expression, head/torso movement) and their alignment to detect prosodic boundaries within each utterance. Based on the differences in the strength of prosodic boundaries, they also analyze syntactic dependencies of clauses. Comparing the utterances produced by a pair of older signers (in their 40 s, from the second generation of deaf people in the Bedouin village where ABSL is found) and those produced by a pair of younger signers (about 15 years younger), the authors have found that younger signers produce sentences with syntactically and prosodically more complex structures than older signers. For example, while prosodic boundaries produced by the older signers are fuzzier, those produced by the younger signers are more systematically marked by intonation and prosodic cues. The differences between the two groups of signers, as the authors claim, mirror the developmental process of the syntactic and prosodic systems of this new language in progress. They also relate the development of the hierarchical structure in the semantic and phonological component in this language to linguistic recursion, which has been claimed to be the only uniquely human component of the faculty of language (Hauser et al., 2002).

New insights: Since thorough documentation of a newly born language and its development is rare, it is not surprising that the contribution by Sandler et al. (as well as related work of this team of researchers) provides us with a range of new insights and many further questions. By looking at the prosodic and syntactic structure of signers from two different age groups, one can find a clear development of the complexity of linguistic expressions. This methodology also allows us to ask fundamental questions about language evolution, such as what are the basic components of the human language, how they develop over time, and in what order.

4. Concluding remarks

This special issue considers two well-established and fundamental aspects of the prosody–syntax interface, namely focus and prosodic phrasing. It also integrates a new and promising line of research: the question about the emergence of the complexity of this interface (subsumed under the notion of language evolution). The eight articles presented in this special issue, partitioned into these three major topics, investigate the interface from different theoretical perspectives, deal with different phenomena, and look at various languages and language families. In addition, the articles represent a variety of methodologies which can be used to investigate topics at the interface. The current issue contains theoretical investigations (Féry), in which proposals are made from a purely theoretical perspective; experimental studies (Kabagema-Bilan et al.; Prieto), in which data are collected through various methodologies such as laboratory recordings or recordings of spontaneous speech; descriptive work based on data collected in fieldwork (Kisseberth and Abasheikh; Sandler et al.); and combinations of these methods (Downing and Mtenje; Feldhausen; Ishihara). We hope that this selection of papers represents a useful overview of the current understanding of and methodological approach to the prosody–syntax interface.

We would like to thank the authors for contributing their work to this issue, and we are grateful to our anonymous colleagues who kindly agreed to review the papers. We are also indebted to the journal editors, Anikó Liptak and Johan Rooryck, for their help and assistance throughout the production process, and most importantly for their interest in our project and for accepting the collection as a special issue of *Lingua*.

References

- Adger, D., 2007. Stress and phasal syntax. *Linguistic Analysis* 33, 238–266.
- Aguilar, L., de-la-Mota, C., Prieto, P. (Eds.), 2009. Cat_ToBI Training Materials. http://prosodia.upf.edu/cat_tobi/ (17 November 2010).

- Boersma, P., Hayes, B., 2001. Empirical tests of the gradual learning algorithm. *Linguistic Inquiry* 32 (1), 45–86.
- Cho, Y.Y., 1990. Syntax and phrasing in Korean. In: Inkelas, S., Zec, D. (Eds.), *The Phonology–Syntax Connection*. The University of Chicago Press, Chicago, pp. 47–62.
- Chomsky, N., 1971. Deep structure, surface structure and semantic interpretation. In: Steinberg, D., Jakobovits, L. (Eds.), *Semantics: An Interdisciplinary Reader in Linguistics, Philosophy and Psychology*. CUP, Cambridge, pp. 183–216.
- Chomsky, N., 2000. Minimalist inquiries: the framework. In: Martin, R., Michaels, D., Uriagereka, J. (Eds.), *Step by Step: In Honor of Howard Lasnik*. MIT Press, Cambridge, MA, pp. 85–155.
- Chomsky, N., Halle, M., 1968. *The Sound Pattern of English*. Harper & Row, New York, NY.
- Dehé, N., 2008. To delete or not to delete: the contexts of Icelandic Final Vowel Deletion. *Lingua* 118 (5), 732–753.
- Dehé, N., 2009. Clausal parentheticals, intonational phrasing, and prosodic theory. *Journal of Linguistics* 45 (3), 569–615.
- D'Imperio, M., Elordieta, G., Frota, S., Prieto, P., Vigário, M., 2005. Intonational phrasing in Romance: the role of syntactic and prosodic structure. In: Frota, S., Vigário, M., Freitas, M.J. (Eds.), *Prosodies: With Special Reference to Iberian Languages*. Mouton de Gruyter, Berlin, pp. 59–97.
- É. Kiss, K., 1998. Identification focus and information focus. *Language* 74, 245–341.
- Feldhausen, I., 2010. Sentential Form and Prosodic Structure of Catalan [Linguistik Aktuell/Linguistics Today 168]. John Benjamins, Amsterdam.
- Féry, C., 2010a. Recursion in Prosodic Structure. *Phonological Studies* (13), 51–60.
- Féry, C., 2010b. Syntax, information structure, embedded prosodic phrasing, and the relational scaling of pitch accents. In: Erteschik-Shir, N., Rochman, L. (Eds.), *The Sound Patterns of Syntax*. Oxford University Press, Oxford, pp. 271–290.
- Féry, C., 2011. Informationsstruktur: Begriffe und grammatische Korrelate. In: *Japanische Gesellschaft für Germanistik* (Ed.), *Mapping zwischen Syntax, Prosodie und Informationsstruktur*. Akten des 37. Linguistenkolloquiums, Kyoto 2009.
- Féry, C., Ishihara, S., 2010. How focus and givenness shape prosody. In: Zimmermann, M., Féry, C. (Eds.), *Information Structure: Theoretical, Typological, and Experimental Perspectives*. OUP, Oxford, pp. 36–63.
- Frota, S., 2002. The prosody of focus: a case study with cross-linguistic implications. *Speech Prosody* 2002, 319–322.
- Ghini, M., 1993. phi-formation in Italian: a new proposal. *Toronto Working Papers in Linguistics* 12, 41–79.
- Gussenhoven, C., 1984. Testing the reality of focus domain. *Language and Speech* 26, 61–80.
- Gussenhoven, C., 1992. Sentence accents and argument structure. In: Roca, I. (Ed.), *Thematic Structure: Its role in Grammar*. Folis, Berlin/New York, pp. 79–106.
- Halliday, M.A.K., 1967. Notes on transitivity and theme in English. Part II. *Journal of Linguistics* 3, 199–244.
- Hauser, M., Chomsky, N., Fitch, T., 2002. The faculty of language: what is it, who has it, and how did it evolve? *Science* 298, 1569–1578.
- Hayes, B., 1995. *Metrical Stress Theory: Principles and Case Theory*. University of Chicago Press, Chicago.
- Hayes, B., Lahiri, A., 1991. Bengali intonational phonology. *Natural Language and Linguistic Theory* 9, 47–96.
- Höhle, T., 1982. Explikation für “normale Betonung” und “normale Wortstellung”. In: Abraham, W. (Ed.), *Satzglieder im Deutschen: Vorschläge zur syntaktischen, semantischen und pragmatischen Fundierung*. Narr, Tübingen, pp. 75–153.
- Hornstein, N., Nunes, J., Grohmann, K., 2005. *Understanding Minimalism*. CUP, Cambridge.
- Ishihara, S., 2003. *Intonation and Interface Conditions*. Ph.D. Dissertation, Massachusetts Institute of Technology.
- Ishihara, S., 2007. Major phrase, focus intonation and multiple spell-out (MaP, FI, MSO). *The Linguistic Review* 24, 137–167.
- Itô, J., Mester, A., 2009. The extended prosodic word. In: Grijzenhout, J., Kabak, B. (Eds.), *Phonological Domains: Universals and Deviations*. Mouton de Gruyter, Berlin, pp. 135–194.
- Itô, J., Mester, A. Recursive Prosodic Phrasing in Japanese. In: Borowsky, T., Kawahara, S., Shinya, T., Sugahara, M. (Eds.), *Prosody Matters: Essays in Honor of Elisabeth Selkirk*. Equinox Publishers, London, forthcoming.
- Jackendoff, R., 1972. *Semantic Interpretation in Generative Grammar*. MIT Press, Cambridge, MA.
- Jun, S.-A. (Ed.), 2005. *Prosodic Typology: The Phonology of Intonation and Phrasing*. OUP, Oxford.
- Jun, S.-A. (Ed.), *Prosodic Typology II*. OUP, Oxford, forthcoming.
- Kahnemuyipour, A., 2004. *The Syntax of Sentential Stress*. Ph.D. dissertation, University of Toronto.
- Kanerva, J.M., 1990. Focus on phonological phrases in Chicheŵa. In: Inkelas, S., Zec, D. (Eds.), *The Phonology–Syntax Connection*. University of Chicago Press, Chicago, pp. 145–161.
- Kisseberth, C.W., 2010. Optimality theory and the theory of phonological phrasing: the Chimwiini evidence. In: Erteschik-Shir, N., Rochman, L. (Eds.), *The Sound Patterns of Syntax*. Oxford University Press, Oxford, pp. 217–246.
- Kratzer, A., Selkirk, E., 2007. Phase theory and prosodic spellout: the case of verbs. *The Linguistic Review* 24, 93–135.
- Krifka, M., 1984. Fokus, Topik, syntaktische Struktur und semantische Interpretation. Manuscript, Munich <<http://amor.rz.hu-berlin.de/~h2816i3x/articles.html>>.
- Krifka, M., 2008. Basic notions of information structure. *Acta Linguistica Hungarica* 55, 243–276.
- Ladd, D.R., 1986. Intonational phrasing: the case of recursive prosodic structure. *Phonology Yearbook* 3, 311–340.
- Ladd, D.R., 1988. Declination ‘reset’ and the hierarchical organization of utterances. *Journal of the Acoustical Society of America* 84, 530–544.
- Ladd, D.R., 2008. *Intonational Phonology*, 2nd ed. CUP, Cambridge.
- Legate, J.A., 2003. Some interface properties of the phase. *Linguistic Inquiry* 34, 506–516.
- Nagahara, H., 1994. *Phonological phrasing in Japanese*. Ph.D. Dissertation, University of California, Los Angeles.
- Nespor, M., Vogel, I., 1986. *Prosodic Phonology*. Foris, Dordrecht.
- Nespor, M., Vogel, I., 2007. *Prosodic Phonology*. With a new Foreword. Mouton de Gruyter, Berlin.
- Pierrehumbert, J., 1980. *The Phonology and Phonetics of English Intonation*. Ph.D. Dissertation, Massachusetts Institute of Technology.
- Pierrehumbert, J., Beckman, M., 1988. *Japanese Tone Structure*. MIT Press, Cambridge, MA.
- Prieto, P. The intonational phonology of Catalan. In: Jun, S.-A. (Ed.), forthcoming.
- Prince, A., Smolensky, P., 2004. *Optimality Theory: Constraint Interaction in Generative Grammar*. Blackwell, Oxford (1993: Rutgers University Center for Cognitive Science Technical Report 2).
- Reinhart, T., 2006. *Interface Strategies: Optimal and Costly Computations*. MIT Press, Cambridge, MA.
- Rooth, M., 1992. A theory of focus interpretation. *Natural Language Semantics* 1, 75–116.
- Samek-Lodovici, V., 2005. Prosody–syntax interaction in the expression of focus. *Natural Languages and Linguistic Theory* 23, 687–755.
- Sandalo, F., Truckenbrodt, H., 2002. Some notes on phonological phrasing in Brazilian Portuguese. In: *MIT Working Papers in Linguistics* 42, pp. 285–310.
- Selkirk, E., 1984. *Phonology and Syntax: The Relation Between Sound and Structure* [Current studies in Linguistics Series 10] MIT Press, Cambridge, MA.
- Selkirk, E., 1986. On derived domains in sentence phonology. *Phonology Yearbook* 3, 371–405.
- Selkirk, E., 1995. The prosodic structure of function words. In: Beckman, J., Dickey, L.W., Urbanczyk, S. (Eds.), *Papers in Optimality Theory*. GLSA Publications, Amherst, (Also in: Morgan, J.L., Demuth, K. (Eds.), 1996, *Signal to Syntax: Bootstrapping from Speech to Grammar in Early Acquisition*. Lawrence Erlbaum Associates, Mahwah, NJ, pp. 187–214), pp. 439–470.
- Selkirk, E., 2000. The interaction of constraints on prosodic phrasing. In: Horne, M. (Ed.), *Prosody: Theory and Experiment*. Kluwer, Dordrecht, pp. 231–262.
- Selkirk, E., 2002. Contrastive FOCUS vs. presentational focus: prosodic evidence from right node raising in English. *Speech Prosody* 2002, 643–646.
- Selkirk, E., 2008. Contrastive focus, givenness and the unmarked status of “discourse-New”. *Acta Linguistica Hungarica* 55, 331–346.
- Selkirk, E., 2009. On clause and intonational phrase in Japanese: the syntactic grounding of prosodic constituent structure. *Gengo Kenkyu* 136, 35–73.
- Selkirk, E., Shen, T., 1990. Prosodic domains in Shanghai Chinese. In: Inkelas, S., Zec, D. (Eds.), *The Phonology–Syntax Connection*. The University of Chicago Press, Chicago, pp. 313–337.
- Selkirk, E., Tateishi, K., 1988. Constraints on Minor Phrase formation in Japanese. *CLS* 24, 316–336.
- Shattuck-Hufnagel, S., Turk, E.A., 1996. A prosody tutorial for investigators of auditory sentence processing. *Journal of Psycholinguistic Research* 25, 193–247.

- Sugahara, M., 2003. Downtrends and post-FOCUS intonation in Japanese. Ph.D. Dissertation, University of Massachusetts, Amherst.
- Trager, G.L., Smith, H.L., 1951. An Outline of English Structure. Battenburg Press, Norman, OK (Reprinted 1957 by American Council of Learned Societies, Washington).
- Truckenbrodt, H., 1995. Phonological Phrases: Their Relation to Syntax, Focus, and Prominence. Ph.D. Dissertation, Massachusetts Institute of Technology.
- Truckenbrodt, H., 1999. On the relationship between syntactic phrases and phonological phrases. *Linguistic Inquiry* 30 (2), 219–255.
- Wagner, M., 2005. Prosody and Recursion. Ph.D. Dissertation, Massachusetts Institute of Technology.
- Xu, Y., 1999. Effects of tone and focus on the formation and alignment of f_0 contours. *Journal of Phonetics* 27, 55–105.
- Zec, D., Inkelas, S., 1990. Prosodically constrained syntax. In: Inkelas, S., Zec, D. (Eds.), *The Phonology–Syntax Connection*. University of Chicago Press, Chicago, pp. 365–378.
- Zubizarreta, M.L., 1998. Prosody, Focus, and Word Order. MIT Press, Cambridge, MA.

Nicole Dehé*
 Ingo Feldhausen
 Shinichiro Ishihara
*University of Hamburg, Institut für Romanistik, Von-Melle-Park 6,
 D-20146, Germany*

*Corresponding author. Tel.: +49 7531882928; fax: +49 7531884149
E-mail address: nicole.dehe@uni-konstanz.de (N. Dehé)